

LAB 11 (FINAL – STRUCTURED & PROFESSIONAL)

Student Vault System (Flask + PostgreSQL + Image Upload + Admin Panel)

Lab Roadmap (Do in this order)

1. Environment setup (Windows + venv)
2. Install libraries
3. Database creation (pgAdmin)
4. Project structure
5. Backend (app.py)
6. Frontend (HTML templates)
7. Styling (style.css)
8. Run & test

1. Theory (Concise but Essential)

1.1 Three-Tier Architecture

Presentation Layer → HTML/CSS (Frontend)

Application Layer → Flask (Backend)

Data Layer → PostgreSQL (Database)

Each layer has a **separate responsibility**:

- Frontend → Display
- Backend → Logic
- Database → Storage

Browser (HTML/CSS)



Flask (Python Backend)



PostgreSQL (Database)

User Input → Flask → Database → Flask → HTML → User

1.2 Image Handling

- Save image **file** in: static/uploads/
- Save only **filename** in DB
- Store image in database (slow, inefficient)
- **Implementation Logic**
 - ✓ **# Save file** file.save("static/uploads/" + filename)
 - ✓ **# Store in DB** image_filename = filename
- Render using(<!-- Display --> html):

1.3 Access Control

- **User view:** no age
- **Admin view:** full data (age + timestamp)

2. Environment Setup (Windows + VS Code)

2.1 Create project & open in VS Code

Lab11_StudentVault

2.2 Create virtual environment

A virtual environment isolates project dependency

Benefits:

- Avoid version conflicts
- Maintain a clean project
- Professional practice

python -m venv venv

2.3 Fix PowerShell script restriction (your case)

Error: running scripts is disabled on this system

Reason: Windows blocks scripts for security

Solution Set-ExecutionPolicy RemoteSigned -Scope CurrentUser

- Allows local scripts (safe)
- Required to activate venv

2.4 Activate environment

```
venv\Scripts\activate
```

2.5 Install packages

```
pip install Flask psycpg2-binary werkzeug
```

2.6 (Optional) Upgrade pip

```
python -m pip install --upgrade pip
```

2.7 Setup All Commands

```
python -m venv venv  
venv\Scripts\activate  
pip install Flask psycpg2-binary werkzeug  
python -m pip install --upgrade pip
```

[3. Database Setup \(pgAdmin 4\)](#)

3.1 Create DB

```
CREATE DATABASE student_vault;
```

3.2 Create table

```
CREATE TABLE profiles (  
    id SERIAL PRIMARY KEY,  
    name VARCHAR(100),  
    major VARCHAR(100),  
    age INT,  
    image_filename VARCHAR(255),  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

Column	Purpose
id	Unique identifier
name	Student name
major	Field of study
age	Sensitive attribute
image_filename	Image reference
created_at	Auto timestamp

4. Project Structure (MANDATORY)

```
Lab11_StudentVault/  
├── static/  
│   ├── css/  
│   │   └── style.css  
│   └── uploads/  
├── templates/  
│   ├── index.html  
│   ├── add_profile.html  
│   ├── students.html  
│   ├── admin_login.html  
│   └── admin.html  
├── app.py  
└── venv/
```

5. Backend – app.py (CORE LOGIC)

Replace password="1234" with your actual PostgreSQL password.

```
from flask import Flask, render_template, request, redirect,  
url_for  
from werkzeug.utils import secure_filename  
import psycopg2  
import os  
  
# 1) Create Flask app FIRST  
app = Flask(__name__)  
app.config['UPLOAD_FOLDER'] = 'static/uploads'  
  
# 2) Admin password (lab-level)  
ADMIN_PASSWORD = "Pgadmin4"  
  
# 3) DB connection  
def get_db_connection():  
    return psycopg2.connect(  
        host="localhost",  
        database="student_vault",  
        user="postgres",  
        password="1234" # ● change if needed  
    )
```

```

# ----- ROUTES -----

# Home (landing page)
@app.route('/')
def index():
    return render_template('index.html')

# Add student
@app.route('/add', methods=['GET', 'POST'])
def add():
    if request.method == 'POST':
        name = request.form['name']
        major = request.form['major']
        age = request.form['age']
        file = request.files['profile_image']

        if file and file.filename != '':
            filename = secure_filename(file.filename)
            file.save(os.path.join(app.config['UPLOAD_FOLDER'],
filename))

            conn = get_db_connection()
            cur = conn.cursor()
            cur.execute(
                "INSERT INTO profiles (name, major, age,
image_filename) VALUES (%s, %s, %s, %s)",
                (name, major, age, filename)
            )
            conn.commit()
            cur.close()
            conn.close()

            return redirect(url_for('students'))

        return render_template('add_profile.html')

# View students (NO age)
@app.route('/students')
def students():
    conn = get_db_connection()
    cur = conn.cursor()
    cur.execute("SELECT id, name, major, image_filename FROM
profiles")
    data = cur.fetchall()
    cur.close()

```

```

        conn.close()

        return render_template('students.html', students=data)

# Admin login
@app.route('/admin', methods=['GET', 'POST'])
def admin():
    if request.method == 'POST':
        password = request.form['password']
        if password == ADMIN_PASSWORD:
            conn = get_db_connection()
            cur = conn.cursor()
            cur.execute("SELECT * FROM profiles")
            data = cur.fetchall()
            cur.close()
            conn.close()
            return render_template('admin.html', students=data)
        else:
            return "Wrong Password ❌"

    return render_template('admin_login.html')

# Run
if __name__ == "__main__":
    os.makedirs(app.config['UPLOAD_FOLDER'], exist_ok=True)
    app.run(debug=True)

```

[6. Frontend Templates](#)

6.1 templates/index.html

```

<!DOCTYPE html>
<html>
<head>
    <title>Student Vault</title>
    <link rel="stylesheet" href="{{ url_for('static',
filename='css/style.css') }}">
</head>
<body>

<div class="home-container">
    <h1>📦 Student Vault System</h1>
    <p class="subtitle">Manage student records

```

efficiently</p>

```
<div class="home-buttons">
  <a href="/add" class="btn">✚ Add Student</a>
  <a href="/students" class="btn">📋 View Students</a>
  <a href="/admin" class="btn admin-btn">🔒 Admin
Panel</a>
</div>
</div>

</body>
</html>
```

6.2 templates/add_profile.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Add Student</title>
</head>
<body>

<h2>Add Student</h2>

<form method="POST" enctype="multipart/form-data">
  <input type="text" name="name" placeholder="Name"
required><br><br>
  <input type="text" name="major" placeholder="Major"
required><br><br>
  <input type="number" name="age" placeholder="Age"
required><br><br>
  <input type="file" name="profile_image" required><br><br>

  <button type="submit">Save</button>
</form>

</body>
</html>
```

6.3 templates/students.html (No Age)

```
<!DOCTYPE html>
<html>
<head>
```

```

        <title>Students</title>
        <link rel="stylesheet" href="{{ url_for('static',
filename='css/style.css') }}">
    </head>
    <body>

    <h2>Students List</h2>

    <div class="grid">
    {% for s in students %}
        <div class="card">
            
            <h3>{{ s[1] }}</h3>
            <p>{{ s[2] }}</p>
        </div>
    {% endfor %}
    </div>

    </body>
</html>

```

6.4 templates/admin_login.html

```

<!DOCTYPE html>
<html>
<head>
    <title>Admin Login</title>
</head>
<body>

    <h2>Admin Login</h2>

    <form method="POST">
        <input type="password" name="password" placeholder="Enter
Password">
        <button type="submit">Login</button>
    </form>

    </body>
</html>

```

6.5 templates/admin.html (Full Data)

```

<!DOCTYPE html>
<html>

```



```

<head>
    <title>Admin Panel</title>
</head>
<body>

<h2>Admin Panel</h2>

<table border="1">
<tr>
    <th>ID</th>
    <th>Name</th>
    <th>Major</th>
    <th>Age</th>
    <th>Image</th>
    <th>Created</th>
</tr>

{% for s in students %}
<tr>
    <td>{{ s[0] }}</td>
    <td>{{ s[1] }}</td>
    <td>{{ s[2] }}</td>
    <td>{{ s[3] }}</td>
    <td></td>
    <td>{{ s[5] }}</td>
</tr>
{% endfor %}

</table>

</body>
</html>

```

7. CSS – static/css/style.css

```

body {
    font-family: 'Segoe UI', sans-serif;
    background: #f4f6f9;
    margin: 0;
    padding: 20px;
}

.home-container {
    height: 90vh;
}

```

```
        display: flex;
        flex-direction: column;
        justify-content: center;
        align-items: center;
    }

    .subtitle {
        color: #666;
        margin-bottom: 30px;
    }

    .home-buttons {
        display: flex;
        gap: 20px;
    }

    .btn {
        padding: 12px 25px;
        background: #007bff;
        color: white;
        border-radius: 6px;
        text-decoration: none;
    }

    .btn:hover {
        background: #0056b3;
    }

    .admin-btn {
        background: #dc3545;
    }

    .grid {
        display: flex;
        gap: 20px;
        flex-wrap: wrap;
        justify-content: center;
    }

    .card {
        background: white;
        padding: 20px;
        width: 230px;
        border-radius: 12px;
    }
```

```
        text-align: center;
    }
}
```

8. Run & Test

```
venv\Scripts\activate
python app.py
```

Open:

<http://127.0.0.1:5000>

9. Test Sequence

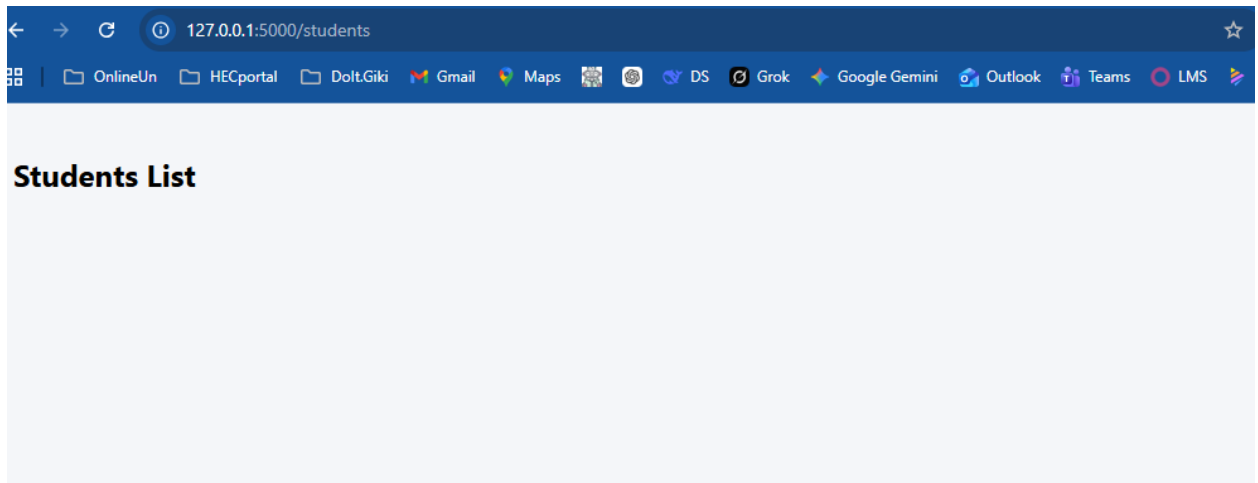
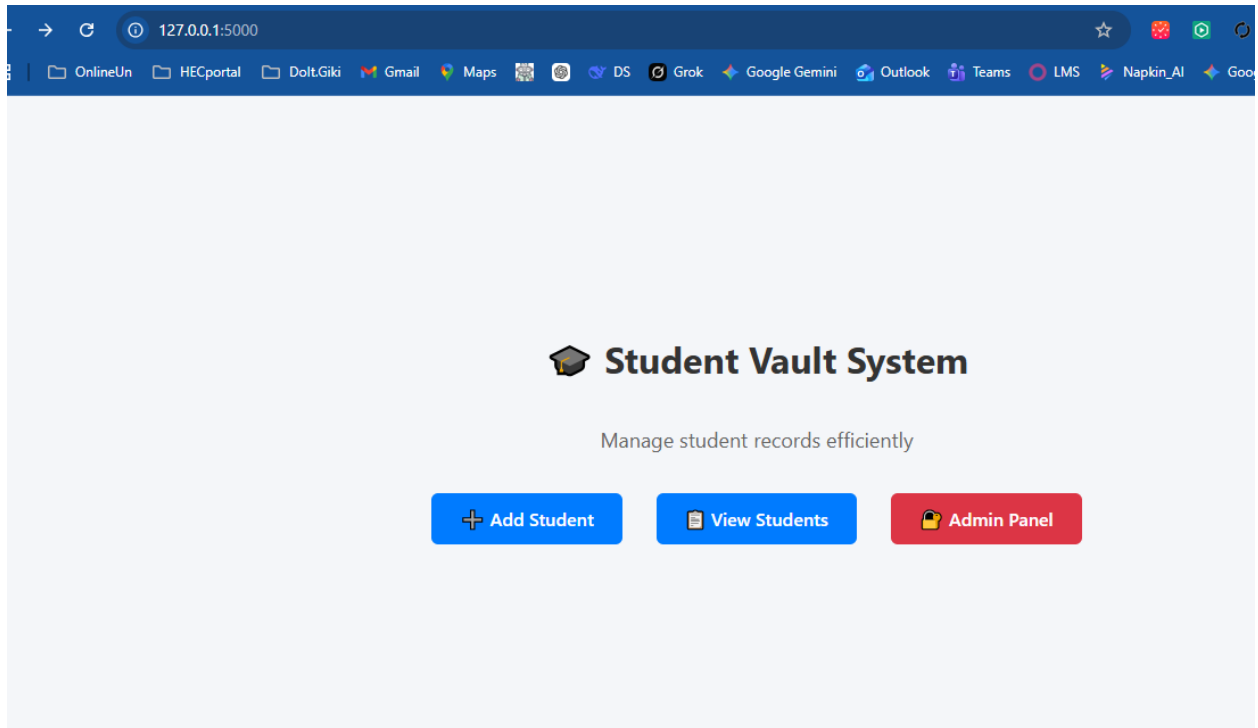
1. Add Student → upload image
2. View Students → no age
3. Admin → password → full data

The screenshot shows the pgAdmin 4 interface. On the left, the 'Object Explorer' pane shows the database structure, with 'Tables' selected under the 'public' schema. The main pane displays a SQL query in the 'Query' tab:

```
1 DROP TABLE IF EXISTS profiles;
2
3 CREATE TABLE profiles (
4     id SERIAL PRIMARY KEY,
5     name VARCHAR(100),
6     major VARCHAR(100),
7     age INT,
8     image_filename VARCHAR(255),
9     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
10 );
11
12 select * from profiles;
```

Below the query, the 'Data Output' tab shows the result of the query. The output is a table with 6 columns: id, name, major, age, image_filename, and created_at. The 'id' column is marked as a primary key (PK) and integer. The 'name', 'major', and 'image_filename' columns are marked as character varying (100), (100), and (255) respectively. The 'age' column is marked as integer. The 'created_at' column is marked as timestamp without time zone.

id	name	major	age	image_filename	created_at
----	------	-------	-----	----------------	------------



Students List



Muhammad Ali
AI



Hassan Raza
AI



Sara Ahmed
Computer and Information Sciences



Bilal Ahmad
CE



Muhammad Sadiq Shah

Admin Panel (All Data)

ID	Name	Major	Age	Image	Created
1	Muhammad Ali	AI	19		2026-04-30 01:27:27.688662
2	Hassan Raza	AI	20		2026-04-30 01:28:28.138949
3	Sara Ahmed	Computer and Information Sciences	17		2026-04-30 01:29:24.582070
4	Bilal Ahmad	CE	20		2026-04-30 01:30:01.035323
5	Muhammad Sadiq Shah	Computer Sciences	23		2026-04-30 01:30:31.634251

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- student_vault
 - Casts
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables**
 - Trigger Functions
 - Types
 - Views
 - Subscriptions
 - Login/Group Roles

student_vault/postgres@PostgreSQL 18*

student_vault/postgres@PostgreSQL 18

Query Query History

```
1 DROP TABLE IF EXISTS profiles;
2
3 CREATE TABLE profiles (
4     id SERIAL PRIMARY KEY,
5     name VARCHAR(100),
6     major VARCHAR(100),
7     age INT,
8     image_filename VARCHAR(255),
9     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
10 );
11
12 select * from profiles;
```

Data Output Messages Notifications

Showing rows: 1 to 5 Page No: 1 of 1

	id [PK] integer	name character varying (100)	major character varying (100)	age integer	image_filename character varying (255)	created_at timestamp without time zone
1	1	Muhammad Ali	AI	19	IMG-20241205-WA0004_1.jpg	2026-04-30 01:27:27.688662
2	2	Hassan Raza	AI	20	ljaz-Hussain-10.png	2026-04-30 01:28:28.138949
3	3	Sara Ahmed	Computer and Information Scien...	17	392928707_852396246892544_8147375410779788904_n...	2026-04-30 01:29:24.58207
4	4	Bilal Ahmad	CE	20	MuhammadSadiqShah_2.jpeg	2026-04-30 01:30:01.035323
5	5	Muhammad Sadiq Shah	Computer Sciences	23	WhatsApp_Image_2024-10-06_at_1.19.23_PM.jpeg	2026-04-30 01:30:31.634251